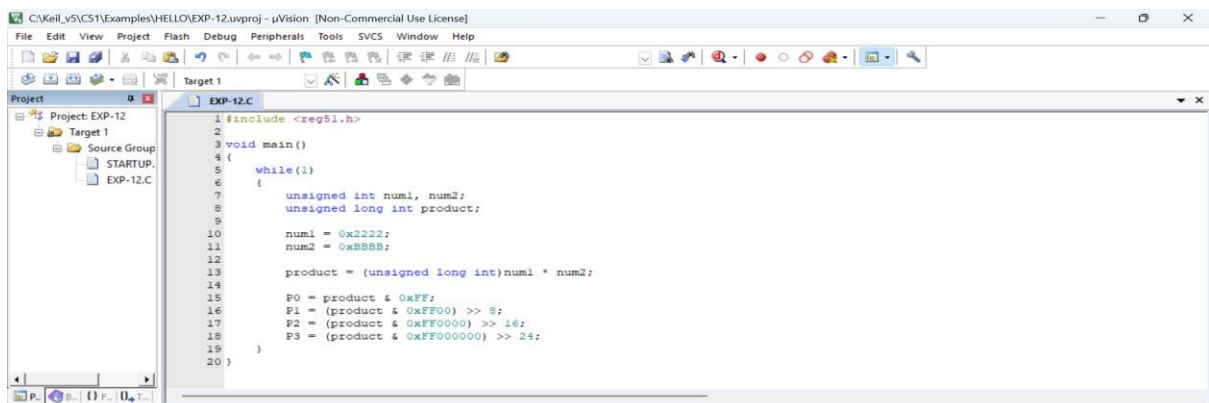
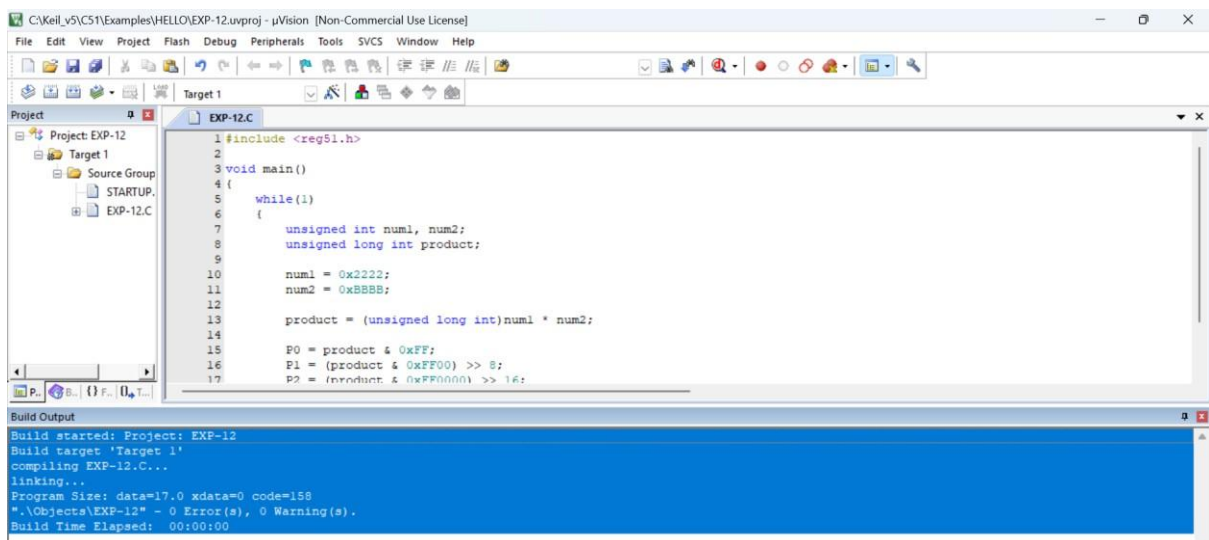


Exp No: 12

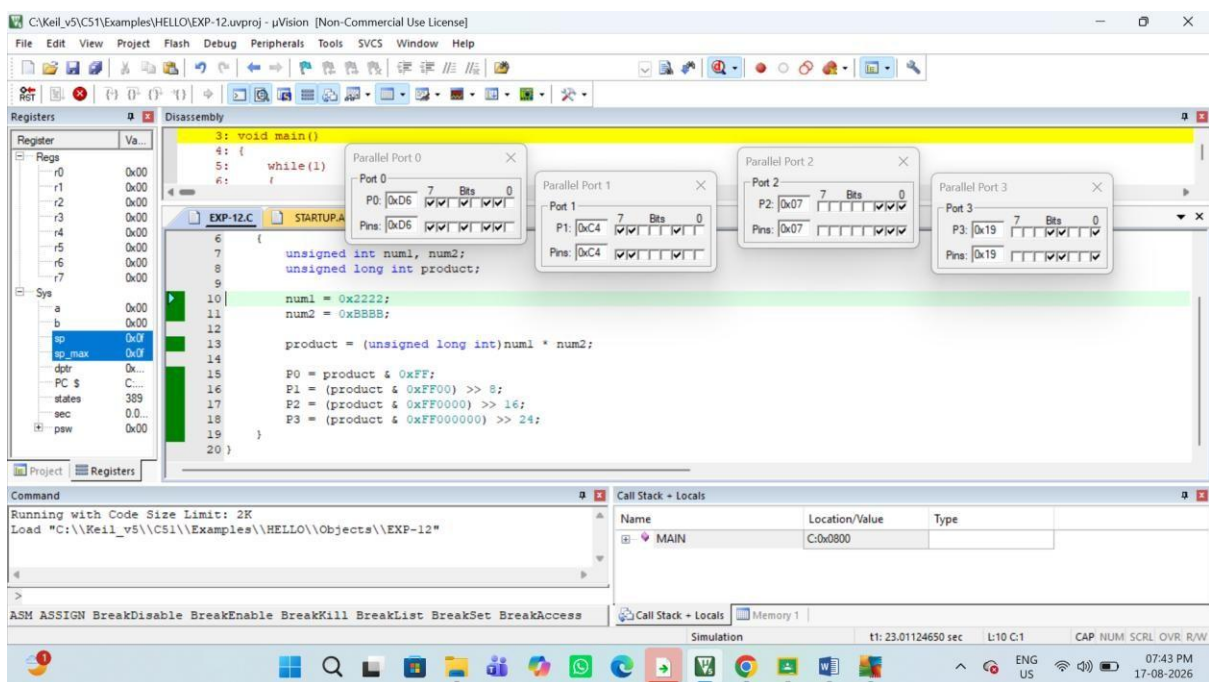
EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS



```
1#include <reg51.h>
2
3void main()
4{
5    while(1)
6    {
7        unsigned int num1, num2;
8        unsigned long int product;
9
10       num1 = 0x2222;
11       num2 = 0xB88B;
12
13       product = (unsigned long int)num1 * num2;
14
15       P0 = product & 0xFF;
16       P1 = (product & 0xFF00) >> 8;
17       P2 = (product & 0xFF0000) >> 16;
18       P3 = (product & 0xFF000000) >> 24;
19
20    }
```



```
Build started: Project: EXP-12
Build target 'Target 1'
compiling EXP-12.C...
linking...
Program Size: data=17.0 xdata=0 code=158
".\Objects\EXP-12" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:00
```



Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
sp	0x00
sp_max	0x00
dptr	0x00
PC	0x00
status	389
sec	0.0...
psw	0x00

Disassembly

```
3: void main()
4: {
5:     while(1)
6:     {
7:         unsigned int num1, num2;
8:         unsigned long int product;
9:
10        num1 = 0x2222;
11        num2 = 0xB88B;
12
13        product = (unsigned long int)num1 * num2;
14
15        P0 = product & 0xFF;
16        P1 = (product & 0xFF00) >> 8;
17        P2 = (product & 0xFF0000) >> 16;
18        P3 = (product & 0xFF000000) >> 24;
19
20    }
```

Hardware Simulation

- Parallel Port 0:** P0: 0x06, Pins: 0x06
- Parallel Port 1:** P1: 0x04, Pins: 0x04
- Parallel Port 2:** P2: 0x07, Pins: 0x07
- Parallel Port 3:** P3: 0x19, Pins: 0x19

Command Window

```
Running with Code Size Limit: 2K
Load "C:\\Keil_v5\\CS1\\Examples\\HELLO\\Objects\\EXP-12"
```

Call Stack - Locals

Name	Location/Value	Type
MAIN	C:0x0800	

Simulation

tt: 23.01124650 sec L:10 C:1 CAP NUM SCRL OVR R/W

07:43 PM 17-08-2026